

ENVS 193DS Spring 2026 Final

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[Link to GitHub repository](#)

Setup

```
# loading packages
library(tidyverse)
library(janitor)
library(gt)
library(here)
library(DHARMA)

# reading in nest data
nests <- read_csv(here("data", "nest_data_final.csv"))
# reading in coffee data
coffee_data <- read_csv(here("data", "coffee_data 2.csv"))
```

Problem 1. Research writing

a.

In part 1, they likely used a logistic regression because they were testing whether distance to the water edge predicts the probability of American Avocet microhabitat use.

In part 2, they likely used a chi-square test of independence because they were testing whether there is a relationship between bird species and habitat type, which are both categorical variables.

b.

For part 1, the probability of American Avocet microhabitat use changed with distance from the water edge, suggesting that distance to water may influence where American Avocets choose to use habitat.

For part 2, American Avocets, Forster's Terns, and Black-necked Stilts used habitats differently, suggesting that each species may have different habitat preferences within the San Francisco Bay.

c.

One additional variable that could influence the probability of American Avocet microhabitat use is water depth (cm), which could be measured at each bird observation location using a meter stick or depth gauge. Water depth likely influences habitat use because American Avocets are wading birds that feed in shallow water, and the paper found that they were commonly associated with wet bare ground and shorelines where water is typically shallower.

d.

Problem 2. Data analysis

a.

```
# cleaning and wrangling nest data
nests_clean <- nests %>%

# selecting columns of interest
select(case_control, height_cm, ant_species) %>%

# keeping only the three ant species needed for analysis
filter(ant_species %in% c("TP", "AB", "RRB")) %>%

# creating a column with full scientific names
mutate(
  ant_species_full = case_when(
    ant_species == "TP" ~ "C. sjostedti",
    ant_species == "AB" ~ "C. mimosae",
    ant_species == "RRB" ~ "C. nigriceps"
  )
) %>%

# reordering ant species levels
mutate(
  ant_species_full = factor(
    ant_species_full,
    levels = c(
      "C. sjostedti",
      "C. mimosae",
      "C. nigriceps"
    )
  )
)
```

```

    )
  )
)

# displaying the structure of the cleaned data frame
str(nests_clean)

tibble [234 x 4] (S3: tbl_df/tbl/data.frame)
 $ case_control      : num [1:234] 1 0 0 0 0 1 0 0 0 0 ...
 $ height_cm         : num [1:234] 392 310 513 154 324 ...
 $ ant_species       : chr [1:234] "RRB" "RRB" "RRB" "RRB" ...
 $ ant_species_full  : Factor w/ 3 levels "C. sjostedti",...: 3 3 3 3 2 3 3 2 3 3 ...

# displaying 10 random rows from the cleaned data frame
slice_sample(nests_clean, n = 10)

```

```

# A tibble: 10 x 4
   case_control height_cm ant_species ant_species_full
   <dbl>      <dbl> <chr>      <fct>
1         0      332. RRB      C. nigriceps
2         1      212. RRB      C. nigriceps
3         0      423  RRB      C. nigriceps
4         0       90  RRB      C. nigriceps
5         0      207  RRB      C. nigriceps
6         0      427  RRB      C. nigriceps
7         0      276. RRB      C. nigriceps
8         0      160. RRB      C. nigriceps
9         1      248. RRB      C. nigriceps
10        0      128  RRB      C. nigriceps

```

b.

In this data set, a value of 1 indicates that a bird nest was present in the tree, while a value of 0 indicates that no bird nest was present. Biologically, this means the response variable represents whether or not a tree was used as a nesting site by birds.

c.

Tree height: Tree height is a continuous variable measured in centimeters. As tree height increases, I would expect the probability of nest occurrence to increase because taller trees

may provide birds with better nesting sites and more protection from predators. I found this information in the Introduction section where the authors discuss how habitat characteristics can influence nesting behavior.

Acacia ant species: Acacia ant species is a categorical variable with three levels: *Crematogaster sjostedti*, *Crematogaster mimosae*, and *Crematogaster nigriceps*. Different ant species may influence nest occurrence because they vary in their behavior and aggressiveness, which could make some trees more attractive or safer for nesting birds than others. I found this information in the Introduction section where the authors describe how species interactions can affect bird nesting behavior.

d.

Model Number	Predictors	Interaction Term?	Model Description
Model 1	Tree height, ant species	No	Tests whether tree height and ant species influence the probability of nest occurrence.
Model 2	Tree height, ant species	Yes	Tests whether the effect of tree height on nest occurrence differs among ant species.

e.

```
# model 1: no interaction
model_1 <- glm(
  case_control ~ height_cm + ant_species_full,
  data = nests_clean,
  family = "binomial"
)

# model 2: interaction model
model_2 <- glm(
  case_control ~ height_cm * ant_species_full,
  data = nests_clean,
  family = "binomial"
)
```

f.

```
# comparing models using AIC
AIC(model_1, model_2)
```

	df	AIC
model_1	4	215.2312
model_2	6	219.1359

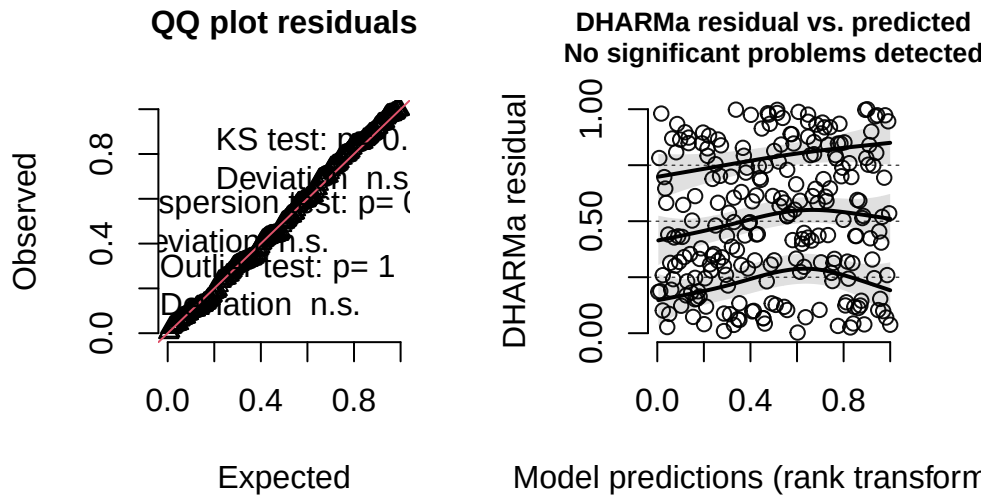
The best model as determined by Akaike's Information Criterion (AIC) included tree height and acacia ant species as predictors of nest occurrence. Including an interaction between tree height and acacia ant species did not improve the model, so the simpler model was selected.

g.

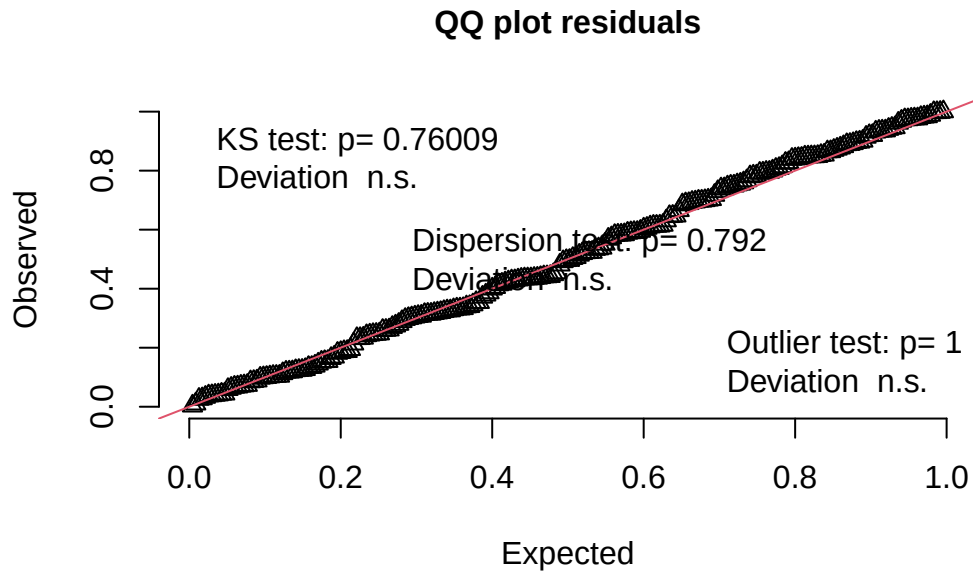
```
# creating simulated residuals for the selected logistic regression model
model_1_residuals <- simulateResiduals(
  fittedModel = model_1
)

# plotting diagnostic results
plot(model_1_residuals)
```

DHARMa residual



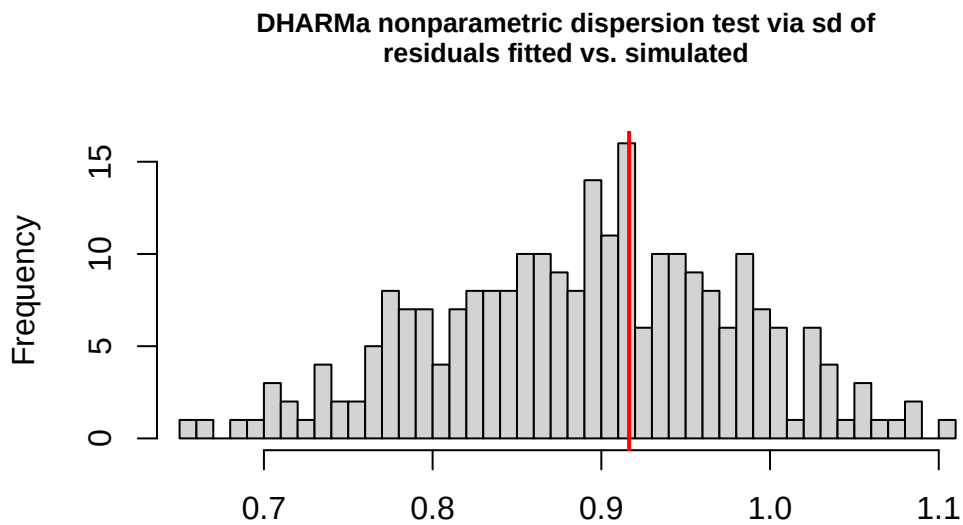
```
# testing for uniformity of residuals
testUniformity(model_1_residuals)
```



Asymptotic one-sample Kolmogorov-Smirnov test

```
data: simulationOutput$scaledResiduals  
D = 0.043813, p-value = 0.7601  
alternative hypothesis: two-sided
```

```
# testing for dispersion  
testDispersion(model_1_residuals)
```

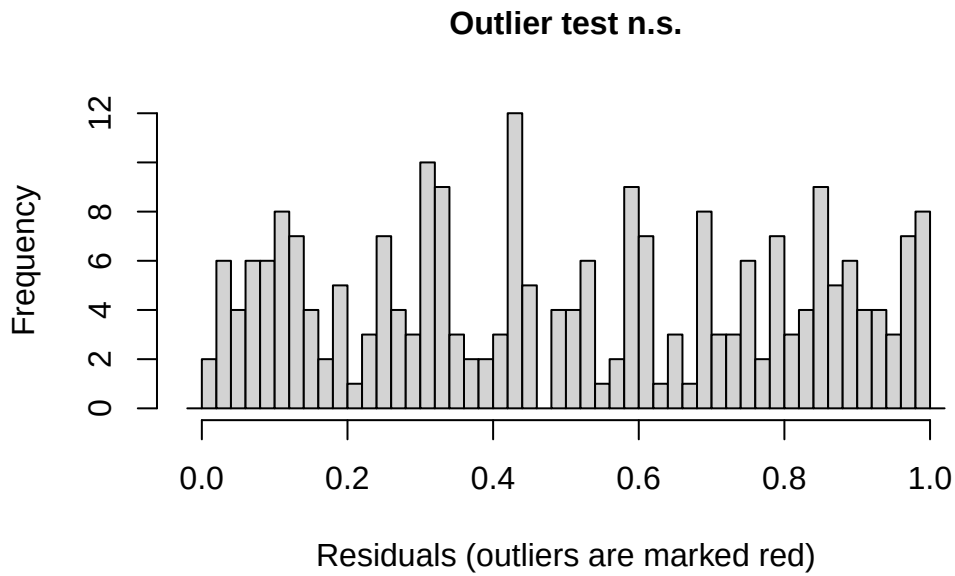


Simulated values, red line = fitted model. p-value (two.sided) = 0.792

DHARMA nonparametric dispersion test via sd of residuals fitted vs.
simulated

```
data: simulationOutput  
dispersion = 1.0292, p-value = 0.792  
alternative hypothesis: two.sided
```

```
# testing for outliers  
testOutliers(model_1_residuals)
```

DHARMA bootstrapped outlier test

```
data: model_1_residuals
outliers at both margin(s) = 0, observations = 234, p-value = 1
alternative hypothesis: two.sided
percent confidence interval:
 0 0
sample estimates:
outlier frequency (expected: 4.27350427350427e-05 )
0
```

The DHARMA diagnostic plots and tests did not show any major issues with the model. The residuals looked fairly even, and there was no evidence of overdispersion or outliers, suggesting that the model fit the data well.

h.

```
# creating prediction data for the selected model
prediction_data <- expand_grid(
```

```

height_cm = seq(
  min(nests_clean$height_cm, na.rm = TRUE),
  max(nests_clean$height_cm, na.rm = TRUE),
  length.out = 100
),
ant_species_full = levels(nests_clean$ant_species_full)
)

# getting model predictions on the logit scale
model_predictions <- predict(
  model_1,
  newdata = prediction_data,
  type = "link",
  se.fit = TRUE
)

# adding predicted probabilities and 95% confidence intervals
prediction_data <- prediction_data %>%
  mutate(
    fit = model_predictions$fit,
    se = model_predictions$se.fit,
    probability = plogis(fit),
    lower_ci = plogis(fit - 1.96 * se),
    upper_ci = plogis(fit + 1.96 * se)
  )

# plotting model predictions with 95% confidence intervals and raw data
ggplot() +

# plotting the underlying data
  geom_jitter(
    data = nests_clean,
    aes(
      x = height_cm,
      y = case_control,
      color = ant_species_full
    ),
    height = 0.05,
    width = 0,
    alpha = 0.4
  ) +

```

```

# adding 95% confidence intervals
geom_ribbon(
  data = prediction_data,
  aes(
    x = height_cm,
    ymin = lower_ci,
    ymax = upper_ci,
    fill = ant_species_full
  ),
  alpha = 0.25
) +

# adding model prediction lines
geom_line(
  data = prediction_data,
  aes(
    x = height_cm,
    y = probability,
    color = ant_species_full
  ),
  linewidth = 1
) +

# using non-default colors
scale_color_manual(
  values = c(
    "C. sjostedti" = "darkgreen",
    "C. mimosae" = "steelblue",
    "C. nigriceps" = "darkorange"
  )
) +
scale_fill_manual(
  values = c(
    "C. sjostedti" = "darkgreen",
    "C. mimosae" = "steelblue",
    "C. nigriceps" = "darkorange"
  )
) +

# putting each ant species in a separate panel
facet_wrap(~ ant_species_full) +

```

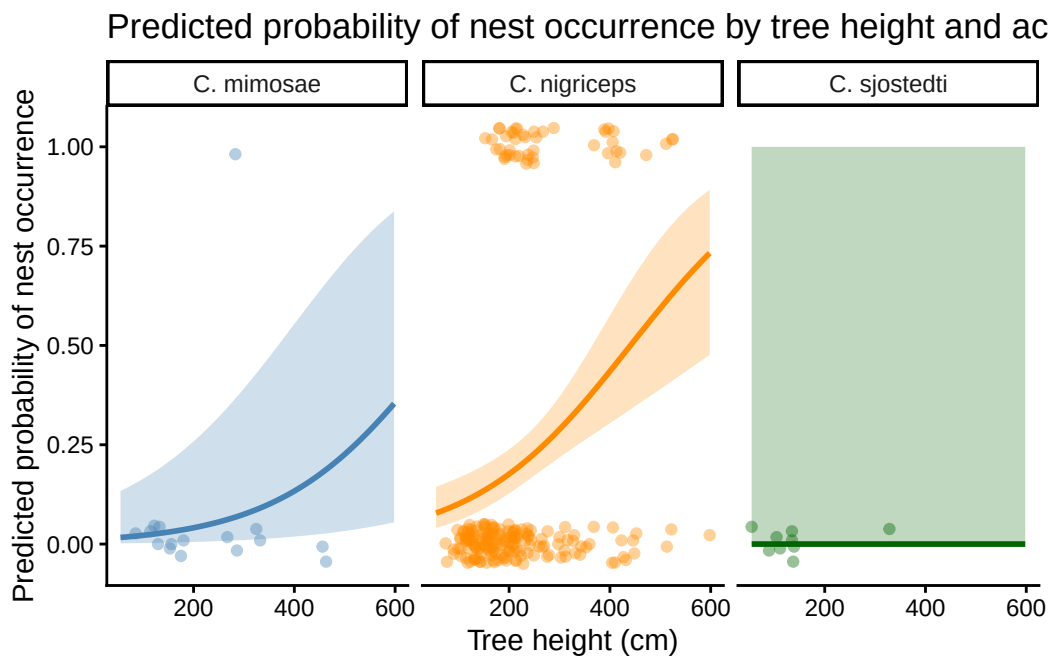
```

# labeling axes and title
labs(
  title = "Predicted probability of nest occurrence by tree height and acacia ant species",
  x = "Tree height (cm)",
  y = "Predicted probability of nest occurrence"
) +

# removing grey background and gridlines
theme_classic() +

# removing the legend
theme(
  legend.position = "none"
)

```



i.

Figure 1. Predicted probability of bird nest occurrence by tree height (cm) and acacia ant species. Points represent individual trees where nests were either present (1) or absent (0). Solid lines represent predicted probabilities of nest occurrence from the selected logistic regression model, while shaded bands represent 95% confidence intervals around the predictions. Separate panels represent trees occupied by *C. mimosae*, *C. nigriceps*, and *C.*

sjostedti. Data source: Lujan, E., Nielsen, R., Short, Z., Wicks, S., Nderitu Watetu, W., Malingati Khasoha, L., Palmer, T. M., Goheen, J. R., & Alston, J. M. 2023. *Data, code, and metadata for: Symbiotic acacia ants drive nesting behavior by birds in an African savanna*. Zenodo. <https://doi.org/10.5281/zenodo.8373322>.

j.

```
# creating new data at a tree height of 600 cm
prediction_600 <- tibble(
  height_cm = 600,
  ant_species_full = c(
    "C. sjostedti",
    "C. mimosae",
    "C. nigriceps"
  )
)

# adding predicted probabilities
prediction_600 <- prediction_600 %>%
  mutate(
    predicted_probability = predict(
      model_1,
      newdata = prediction_600,
      type = "response"
    )
  )

# displaying results
prediction_600
```

```
# A tibble: 3 x 3
  height_cm ant_species_full predicted_probability
    <dbl> <chr>                <dbl>
1     600 C. sjostedti          0.000000418
2     600 C. mimosae           0.356
3     600 C. nigriceps          0.735
```

k.

Figure 1 shows that the probability of nest occurrence generally increased with tree height for *C. mimosae* and *C. nigriceps*, but remained very low for *C. sjostedti*. At a tree height of 600 cm, the predicted probability of nest occurrence was 0.356 for *C. mimosae*, 0.735 for *C. nigriceps*, and essentially 0 for *C. sjostedti*. These results suggest that both tree height and acacia ant species influence the probability of bird nest occurrence. One possible explanation is that birds may prefer nesting in trees occupied by less aggressive ant species, while avoiding trees occupied by more aggressive ants that could disturb nests.

Problem 3. Affective and exploratory visualizations

a.

How are the visualizations different from each other in the way you have represented your data?

My exploratory visualizations from Homework 2 used standard graphs in R to show relationships between variables using points and axes. In contrast, my affective visualization represents the data as a garden, where flower height represents hours slept, flowers represent whether I exercised, roots represent schoolwork hours, and leaves represent cups of coffee. This makes the data feel more personal and visually engaging than a traditional graph.

What similarities do you see between all your visualizations?

All of my visualizations are based on the same personal data that I collected about sleep, coffee consumption, exercise, and schoolwork. They all help communicate patterns in my daily habits, but they do so in different ways. Even though the affective visualization is more artistic, it still represents the same variables shown in my exploratory graphs.

What patterns do you see in each visualization? Are these different between visualizations? If so, why? If not, why not?

In my Homework 2 scatterplot, I found a weak negative relationship between sleep and coffee consumption, meaning I generally drank more coffee when I slept less. My exercise visualization showed no clear relationship between exercise and sleep. The affective visualization shows similar patterns, but instead of focusing on statistical relationships, it emphasizes how different parts of my daily routine contribute to my overall energy and well-being.

What kinds of feedback did you get during week 9 in workshop or from the instructors? How did you implement or try those suggestions?

I received feedback from peers that my visualization was creative but could be improved by making the symbols easier to interpret. I added a clearer key and made the different parts

of the flowers represent specific variables more consistently. These changes helped make the visualization easier to understand while still keeping the artistic style that I wanted.

b.

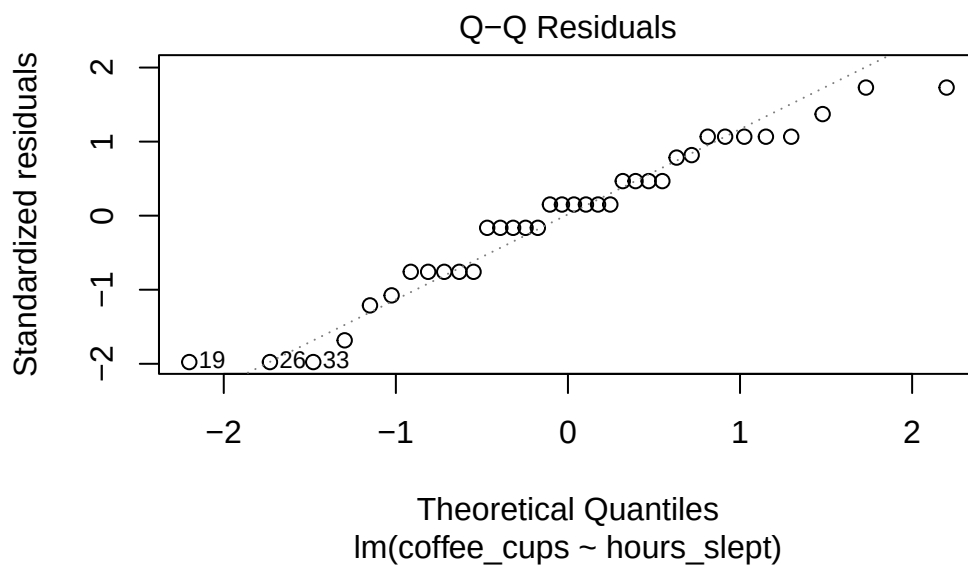
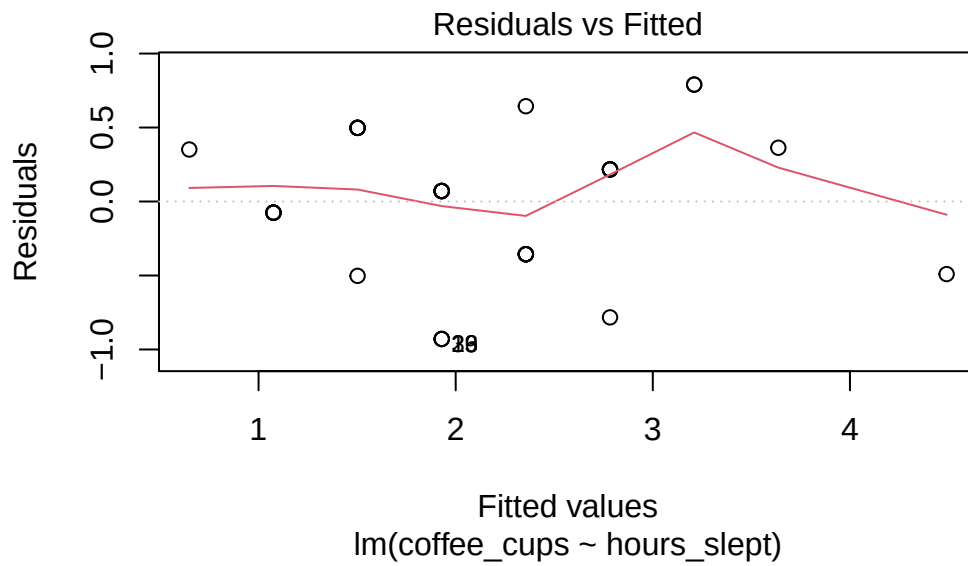
One analysis I could use is a simple linear regression. My research question focuses on whether the amount of sleep I get influences how much coffee I drink, so a linear regression would help test whether there is a relationship between these two variables. This analysis is appropriate because hours slept is a continuous predictor variable measured in hours, and coffee consumption is a continuous response variable measured in cups per day.

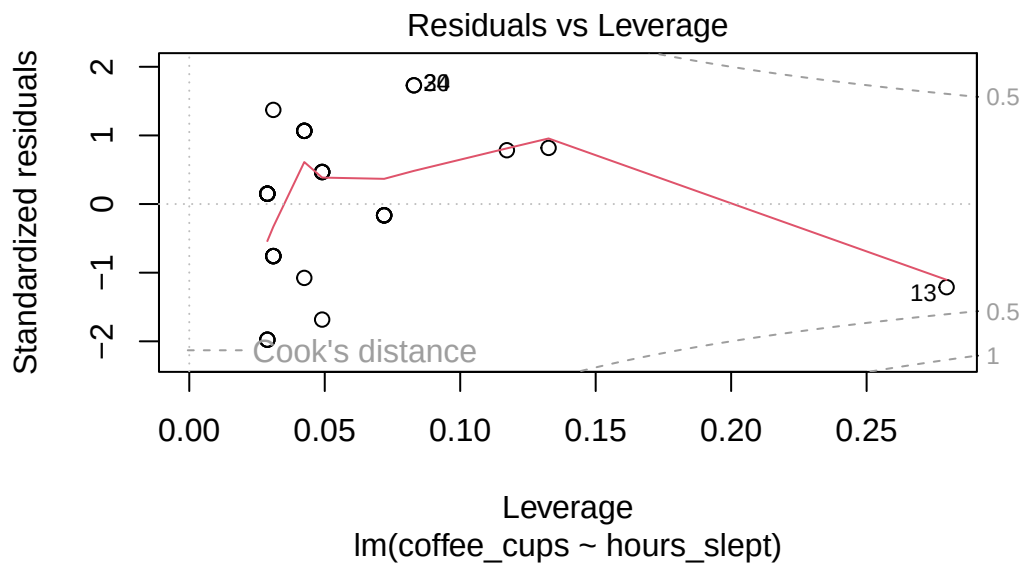
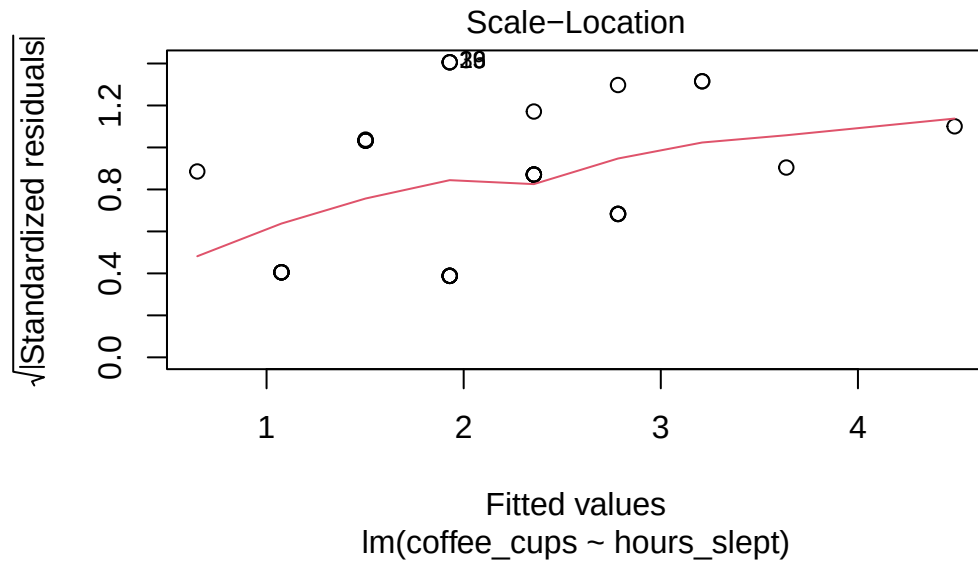
c.

Checking assumptions

```
# creating a linear model to test the relationship between sleep and coffee consumption
coffee_lm <- lm(
  coffee_cups ~ hours_slept,
  data = coffee_data
)

# creating diagnostic plots to check assumptions
plot(coffee_lm)
```





I checked the assumptions of linear regression using diagnostic plots. I examined the residuals for linearity, normality, equal variance, and influential observations. Based on the diagnostic plots, the assumptions appeared reasonable, so I proceeded with the linear regression analysis

Running the analysis

```
# running a simple linear regression
summary(coffee_lm)
```

Call:

```
lm(formula = coffee_cups ~ hours_slept, data = coffee_data)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.92918	-0.35606	0.07082	0.35441	0.79019

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	7.90543	0.58425	13.53	3.02e-15 ***
hours_slept	-0.85375	0.08488	-10.06	1.00e-11 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4769 on 34 degrees of freedom

Multiple R-squared: 0.7485, Adjusted R-squared: 0.7411

F-statistic: 101.2 on 1 and 34 DF, p-value: 1.004e-11

To test whether sleep influences coffee consumption, I used a simple linear regression with coffee cups as the response variable and hours slept as the predictor variable. The analysis showed a significant negative relationship between sleep and coffee consumption ($F(1, 34) = 101.2$, $p < 0.001$). As the number of hours slept increased, the number of cups of coffee consumed tended to decrease.

Effect size

```
# calculating R-squared as an effect size
summary(coffee_lm)$r.squared
```

```
[1] 0.7484771
```

The effect size (R^2) was 0.748. This means that about 74.8% of the variation in coffee consumption can be explained by the number of hours slept, indicating a strong relationship between these variables.

d.

```
# creating a visualization of the linear regression model
ggplot(
  coffee_data,
  aes(
    x = hours_slept,
    y = coffee_cups
  )
) +

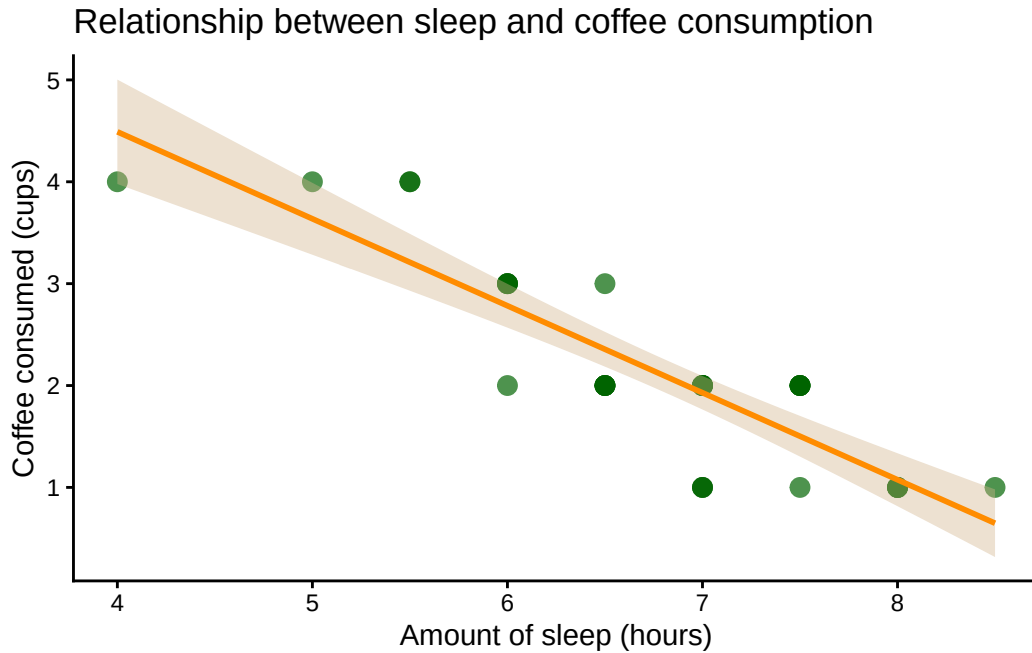
# showing the underlying data
geom_point(
  color = "darkgreen",
  size = 3,
  alpha = 0.7
) +

# adding model prediction line with 95% confidence interval
geom_smooth(
  method = "lm",
  se = TRUE,
  color = "darkorange",
  fill = "tan"
) +

# changing the theme
theme_classic() +

# labeling the plot and axes
labs(
  title = "Relationship between sleep and coffee consumption",
  x = "Amount of sleep (hours)",
  y = "Coffee consumed (cups)"
)
```

`geom\_smooth()` using formula = 'y ~ x'



e.

Figure 2. Relationship between sleep and coffee consumption. Green points represent individual daily observations of hours slept and cups of coffee consumed. The orange line represents the predicted relationship from a simple linear regression model, while the shaded tan area represents the 95% confidence interval around the prediction. The figure shows a negative relationship between sleep and coffee consumption, with coffee intake generally decreasing as hours of sleep increase. Data source: Personal dataset collected by Elena Donelson from 2026-04-17 to 2026-05-23.

f.

I found a significant negative relationship between hours slept and coffee consumption. As the amount of sleep increased, the number of cups of coffee consumed generally decreased ($F(1, 34) = 101.2, p < 0.001, R^2 = 0.748$). This relationship was fairly strong, with hours slept explaining about 74.8% of the variation in coffee consumption. Figure 2 shows this trend, with higher coffee consumption generally occurring on days when I got less sleep.

g.